

- 2 Trypsin is a protease enzyme. A student compared the activity of trypsin from two different species: the Atlantic salmon, *Salmo salar*, and the domestic pig, *Sus scrofa domesticus*.

Fig. 2.1 shows an Atlantic salmon and Fig. 2.2 shows domestic pigs.



Fig. 2.1



Fig. 2.2

Atlantic salmon are fish and are ectothermic, which means that the internal body temperature fluctuates with the surrounding environmental temperature.

The water temperature in the habitat of Atlantic salmon can decrease to -0.5°C .

Domestic pigs are mammals and maintain an internal body temperature of approximately 38°C .

The student was provided with trypsin from Atlantic salmon, trypsin from domestic pigs and cubes of gelatine. Gelatine is made from the protein collagen.

The student measured the time taken for the trypsin from each animal species to break down the gelatine cubes at 20 °C.

The student:

- used 10cm³ of 5% trypsin solution from each animal species
- placed each trypsin solution in separate test-tubes, maintained at 20 °C in a water-bath
- placed a gelatine cube in each test-tube of trypsin solution
- recorded the time taken to break down the gelatine cube in each test-tube.

The student repeated this 12 times for each type of trypsin.

(a) (i) Identify the independent variable **and** the dependent variable in this experiment.

independent variable

dependent variable

[2]

(ii) The student standardised the temperature of the water-bath, the volume of the trypsin solution and the concentration of the trypsin solution.

State **one** other variable that the student should standardise in this experiment.

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..... [1]

- (b) The results of the experiment are shown in Table 2.1.

Table 2.1

replicate	time taken to break down gelatine cube/seconds	
	Atlantic salmon	domestic pigs
1	125	130
2	121	127
3	128	134
4	119	128
5	132	135
6	125	130
7	131	132
8	127	128
9	126	128
10	131	137
11	131	133
12	129	131
mean	127.1	
standard deviation (<i>s</i>)	4.1	

- (i) Use the data in Table 2.1 to calculate the mean time taken for trypsin from domestic pigs to break down the gelatine cube.

mean = seconds [1]

- (ii) The formula for calculating standard deviation is:

$$s = \sqrt{\frac{\sum(x - \bar{x})^2}{n - 1}}$$

Key to symbols

x = observation

$\bar{x}$ = mean

n = sample size (number of observations)

The student calculated $\sum(x - \bar{x})^2$ as **110.92** for the domestic pigs.

Use the calculated value of $\sum(x - \bar{x})^2$ and the data in Table 2.1 to complete the calculation of the standard deviation for domestic pigs.

s = seconds [1]

- (iii) The student used a t -test to analyse the data.

The null hypothesis for this t -test was:

There is no difference between the time taken for trypsin from Atlantic salmon and trypsin from domestic pigs to break down a gelatine cube at 20°C.

The calculated value of t was **2.663**.

The student compared **2.663** to the values in Table 2.2.

Table 2.2

degrees of freedom	probability level (p)			
	0.10	0.05	0.01	0.001
20	1.725	2.086	2.845	3.850
21	1.721	2.080	2.831	3.819
22	1.717	2.074	2.819	3.792
23	1.714	2.069	2.807	3.767
24	1.711	2.064	2.797	3.745

Using Table 2.1, Table 2.2 and the calculated value of t of **2.663**, state **and** explain what the student can conclude about the results.

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- (iv) Suggest explanations for the difference between the activity of the enzyme trypsin in Atlantic salmon and the activity of trypsin in domestic pigs.

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[Total: 10]